

Example 3.27. 1. Consider the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

2. Consider the series $\sum_{n=1}^{\infty} 2^n$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

3. Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \rightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Example 3.28. Show that the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges.

Proof. Let $a_n = \frac{2^n n!}{n^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2 \left(\frac{n}{n+1} \right)^n.$$

Since $(1 + \frac{1}{n})^n \rightarrow e$, we have

$$\left(\frac{n}{n+1} \right)^n = \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{1}{e}.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges. □

Theorem 3.29 (Gauss's Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. Suppose $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^p}$, where $p > 1$ and $\{b_n\}$ is a bounded sequence. Then $\sum_{n=1}^{\infty} a_n$ converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.

Example 3.30. Test the convergence of the series

$$1 + \frac{2^2}{3^2} + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2} + \frac{2^2 \cdot 4^2 \cdot 6^2}{3^2 \cdot 5^2 \cdot 7^2} + \cdots$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. For $n \geq 2$, $a_n = \frac{2^2 \cdot 4^2 \cdots (2n-2)^2}{3^2 \cdot 5^2 \cdots (2n-1)^2}$. Then

$$\frac{a_n}{a_{n+1}} = \frac{(2n+1)^2}{(2n)^2} = \frac{4n^2 + 4n + 1}{4n^2} = 1 + \frac{1}{n} + \frac{1}{4n^2}.$$

Thus $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^2}$, where $\alpha = 1$ and $b_n = \frac{1}{4}$ is bounded. Hence, by Gauss's Test, the series diverges. $\square$

Definition 3.6. Let $f, \phi : \mathbb{N} \rightarrow \mathbb{R}$, where $\phi(n) > 0$ for all sufficiently large n . We write

$$f(n) = O(\phi(n))$$

if there exist constants $k > 0$ and $N \in \mathbb{N}$ such that

$$|f(n)| \leq k \phi(n), \quad \forall n \geq N.$$

In other words, for sufficiently large n , the function $f(n)$ does not grow faster than a constant multiple of $\phi(n)$. In particular, $O(1)$ denotes the class of all bounded functions.

Example 3.31. Show that the polynomial $f(n) = 5n^2 + 3n + 1$ is of order $O(n^2)$.

Proof. For $n \geq 1$, $3n \leq 3n^2$, $1 \leq n^2$. Hence,

$$f(n) = 5n^2 + 3n + 1 \leq 5n^2 + 3n^2 + n^2 = 9n^2.$$

Thus, $|f(n)| \leq 9n^2$, $\forall n \geq 1$. Therefore $f(n) = O(n^2)$. $\square$

Theorem 3.32 (Alternative Form of Gauss's Test). Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. If

$$\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + O\left(\frac{1}{n^p}\right), \quad p > 1,$$

then the series converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.

Example 3.33. Test the convergence of the series

$$\left(\frac{1}{2}\right)^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then $a_n = \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n}\right)^2$. Hence,

$$\frac{a_n}{a_{n+1}} = \left(\frac{2n+2}{2n+1}\right)^2 = \left(1 + \frac{1}{n}\right)^2 \left(1 + \frac{1}{2n}\right)^{-2}.$$

Using the expansion

$$\left(1 + \frac{1}{2n}\right)^{-2} = 1 - \frac{1}{n} + O\left(\frac{1}{n^2}\right),$$

we obtain $\frac{a_n}{a_{n+1}} = 1 + \frac{1}{n} + O\left(\frac{1}{n^2}\right)$. Therefore, $\alpha = 1$, and by the alternative form of Gauss's Test, the series diverges. $\square$

Exercise 3.2. Test the convergence of the following series.

1.

$$1 + \frac{1}{2}x + \frac{2^2}{3!}x^2 + \frac{3^2}{4!}x^3 + \cdots, \quad x > 0.$$

2.

$$\frac{3}{7} + \frac{3 \cdot 6}{7 \cdot 10}x + \frac{3 \cdot 6 \cdot 9}{7 \cdot 10 \cdot 13}x^2 + \cdots, \quad x > 0.$$

3.

$$\frac{1+x}{1!} + \frac{(1+2x)^2}{2!} + \frac{(1+3x)^3}{3!} + \cdots, \quad x > 0.$$

4.

$$1 + \frac{2^2}{3^2}x + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2}x^2 + \cdots, \quad x > 0.$$

5.

$$1 + \frac{1}{2!}x + \frac{(2!)^2}{4!}x^2 + \frac{(3!)^2}{6!}x^3 + \cdots, \quad x > 0.$$

6.

$$\frac{a}{b} + \frac{a(a+c)}{b(b+c)} + \frac{a(a+c)(a+2c)}{b(b+c)(b+2c)} + \cdots, \quad a, b, c > 0.$$

3.3 Tests of convergence for Alternating Series

Definition 3.7. A series of the form $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ where (a_n) is a sequence of positive terms is called an alternating series.

Definition 3.8 (Absolutely Convergent Series). An infinite series $\sum_{n=1}^{\infty} a_n$ is said to be **absolutely convergent** if the series

$$\sum_{n=1}^{\infty} |a_n|$$

is convergent.

Theorem 3.34. An absolutely convergent series is convergent.

Proof. Let $\sum_{n=1}^{\infty} u_n$ be a series of positive and negative real numbers and suppose that it is absolutely convergent. Then

$$\sum_{n=1}^{\infty} |u_n|$$

is a convergent series of positive terms.